Alias: Corbato

Homework 4 MultiCycle B

stw regx, offset(regy)

- Stores regx into MEM[offset+regy] and MEM[offset+regy+4]
- Sets regy to regy+4

31 26	25 21	20 16	15 0
…	$regy	$regx	offset

Datapath elements:

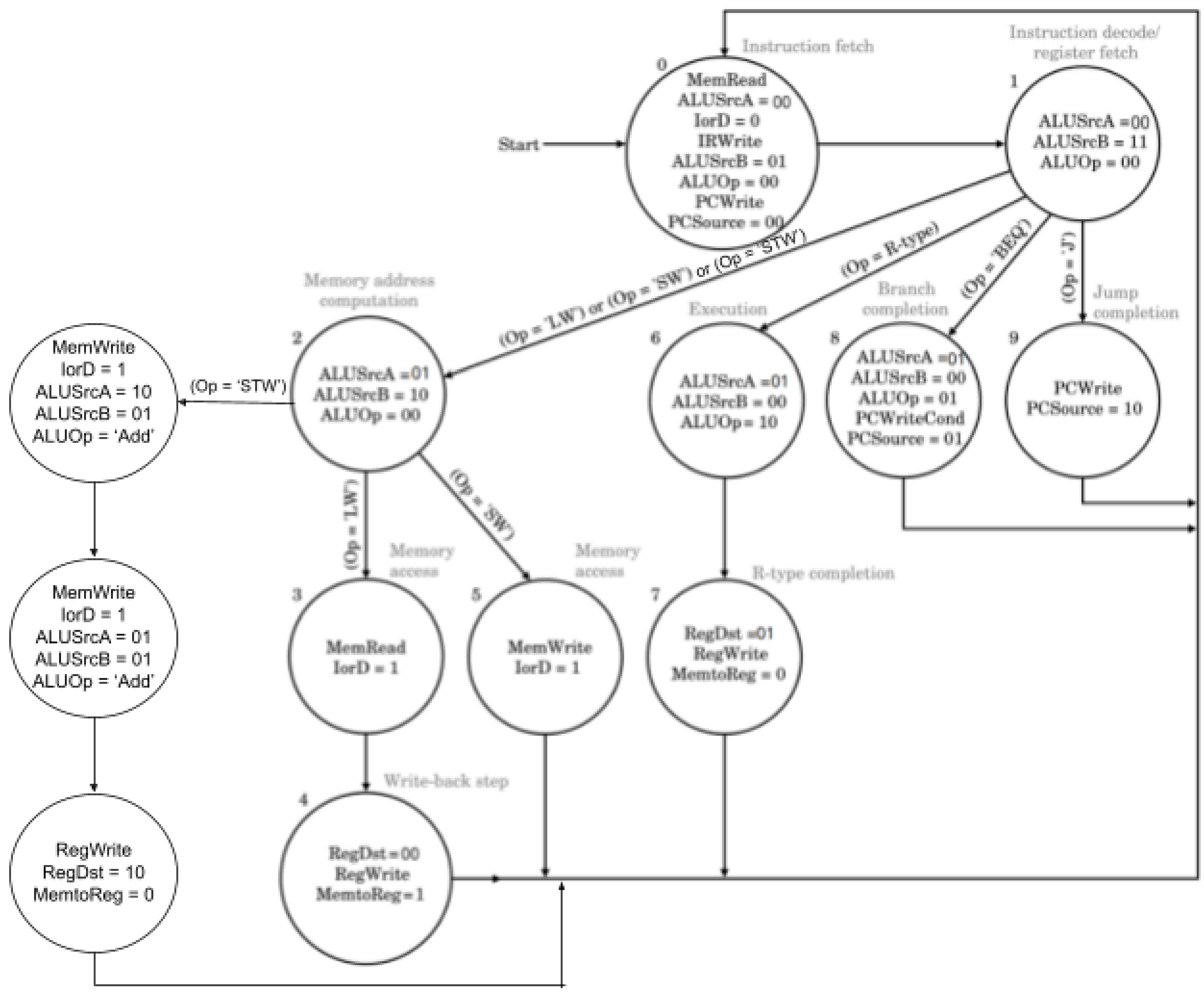
1. PC, Mem, IR (fetch)
2. RF, A, B, sign ext. for immediate (fetch)
3. ALU
4. ALUOut

RTL:

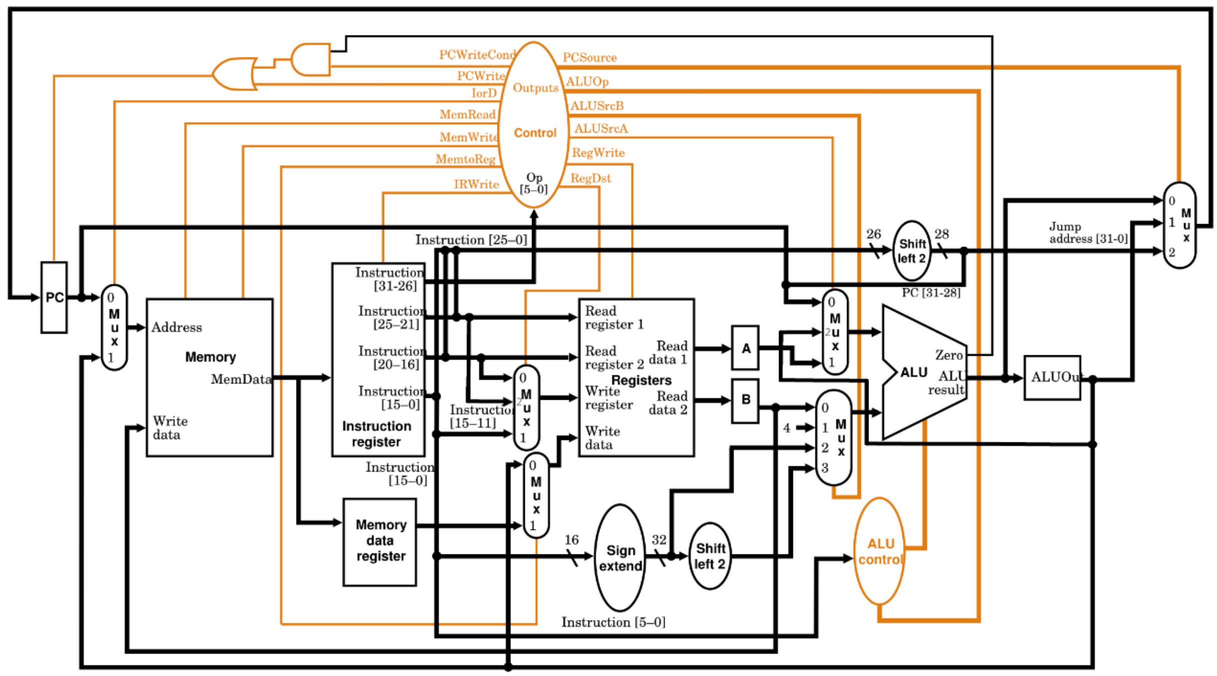
- Fetch
- Decode
- Execute
    1. ALUOut <- A + signext(IR[15:0])
       B <- RF[IR[20:16]]
    2. MEM[ALUOut] <- B
       ALUOut <- ALUOut + 4
       B <- RF[IR[20:16]]
       A <- RF[IR[25:21]]
    3. MEM[ALUOut] <- B
       ALUOut <- A+4
    4. RF[IR[25:21]] <- ALUOut

Datapath changes:

1. 3:1 MUX to RF Write register - 2 connects from IR[25-21]
2. 3:1 MUX to ALU A - 2 connects to ALUOut.



Instruction fetch
0
MemRead
ALUSrcA = 00
IorD = 0
IRWrite
ALUSrcB = 01
ALUOp = 00
PCWrite
PCSource = 00
Start
Instruction decode/
register fetch
1
ALUSrcA =00
ALUSrcB = 11
ALUOp = 00
(Op = 'LW') or (Op = 'SW') or (Op = 'STW')
(Op = R-type)
(Op = 'BEQ')
(Op = 'J')
Memory address
computation
2
ALUSrcA =01
ALUSrcB = 10
ALUOp = 00
Execution
6
ALUSrcA =01
ALUSrcB = 00
ALUOp = 10
Branch
completion
8
ALUSrcA =01
ALUSrcB = 00
ALUOp = 01
PCWriteCond
PCSource = 01
Jump
completion
9
PCWrite
PCSource = 10
(Op = 'STW')
MemWrite
IorD = 1
ALUSrcA = 10
ALUSrcB = 01
ALUOp = 'Add'
MemWrite
IorD = 1
ALUSrcA = 01
ALUSrcB = 01
ALUOp = 'Add'
RegWrite
RegDst = 10
MemtoReg = 0
(Op = 'LW')
(Op = 'SW')
Memory
access
Memory
access
3
MemRead
IorD = 1
5
MemWrite
IorD = 1
R-type completion
7
RegDst =01
RegWrite
MemtoReg = 0
Write-back step
4
RegDst = 00
RegWrite
MemtoReg = 1


PCWriteCond
PCWrite
IorD
MemRead
MemWrite
MemtoReg
IRWrite
Outputs
Control
Op [5–0]
PCSource
ALUOp
ALUSrcB
ALUSrcA
RegWrite
RegDst
PC
Mux
Address
Memory
MemData
Write data
Instruction [31-26]
Instruction [25–21]
Instruction [20–16]
Instruction [15–0]
Instruction register
Instruction [15–0]
Memory data register
Instruction [25–0]
Instruction [15–11]
Read register 1
Read register 2
Registers
Write register
Write data
Read data 1
Read data 2
A
B
4
16
Sign extend
32
Shift left 2
Instruction [5–0]
26
28
PC [31-28]
ALU
Zero
ALU result
ALUOut
ALU control
Jump address [31-0]